Minn Bo Bo

Software Engineer & Researcher | Digital Twins, IoT & Cyber-Physical Systems

Bangkok, Thailand
+66 803791601
minnbobo.official@gmail.com
linkedin.com/in/minn-bo-bo-6b30b0187
github.com/boboTheDev

RESEARCH PROFILE

Electrical and Computer Engineering researcher and software engineer with an MEng GPA of 4.00/4.00, combining IoT and cloud experience with end-to-end development of real-time virtual-world and digital-twin infrastructure. Primary developer of the MANGOs ecosystem, including its learning platform and browser-based Scene Studio, with experience translating research concepts into deployed systems.

Research interests: Digital twins; cyber-physical systems; IoT and sensor-to-cloud platforms; UWB-enabled sensing and localization; real-time systems; immersive simulation; edge-cloud integration; applied AI.

TECHNICAL SKILLS

IoT & Real-Time: Arduino, microcontrollers, sensor-to-cloud workflows, WebSockets, OAuth/SSO, real-time sessions, REST APIs, UWB/IoT research

Digital-Twin Platforms: scene manifests, GLB asset workflows, versioned publishing, interactions, Worlds, canvases, plots, storage

Software: TypeScript, JavaScript, Python, C++, Node.js, Express.js, Flask, React, Redux Toolkit

Cloud & DevOps: AWS, EC2, S3, IAM, CloudWatch, Docker, Docker Compose, Nginx, Linux, GitHub Actions, CI/CD

Data: PostgreSQL, MongoDB, Microsoft SQL Server, Sequelize ORM, SQL, structured data models

AI & Testing: TensorFlow, TensorFlow.js, deep learning, Jest, unit testing, integration testing

PROFESSIONAL EXPERIENCE

Software Engineer - Full Stack & DevOps
Aug 2024 - Present

Terrafirma | Software delivery for William Green, Australia

- Develop and maintain production full-stack systems for an Australian dental equipment distribution and servicing company.
- Build TypeScript and Node.js backend services, REST APIs, and PostgreSQL data workflows.
- Support delivery with AWS, Docker Compose, Nginx, Linux, GitHub Actions, CI/CD, Jest, unit testing, and integration testing.

Full-Stack Developer & DevOps Engineer
Oct 2023 - Aug 2024

iBeky | Medical services technology

- Led Node.js backend development for internal web and mobile projects using Express.js and Jest.
- Provisioned AWS environments, configured Nginx reverse proxies, and administered databases and disaster-response procedures.
- Built Python utilities and Flask applications for AI-enabled document and data workflows.

Junior Full-Stack Developer
Jan 2023 - Oct 2023

Jonda Health | Singapore healthcare technology

- Served as principal backend developer for JondaX, guided interns, developed Python document-processing tools, and integrated AWS services.

Backend Developer & Database Designer
2021 - 2022

Australian learning management system startup

Designed database systems, secure routes, permission models, and REST-based Azure Functions with Node.js, Express.js, Sequelize ORM, and JavaScript.

EDUCATION

Master of Engineering, Electrical Engineering | GPA: 4.00/4.00 **2024 - 2026**
Chulalongkorn University, Thailand | Telecommunication and Networking Systems

BEng (Hons), Electrical and Electronic Engineering - First Class **2022 - 2024**
Liverpool John Moores University, United Kingdom

Electronic and Communication Engineering **2016 - 2022**
Yangon Technological University, Myanmar | Transferred

SELECTED RESEARCH & ENGINEERING SYSTEM

MANGOs Scene Studio & Educational Metaverse Ecosystem - Primary Developer
MANGOs | School | Scene Studio | Chulalongkorn University
Mentored by Professor Lunchakorn Wuttisittikulkij

- Built the full browser-based Scene Studio software platform: scene manifests, asset ingestion, draft and release versioning, camera and interaction models, decoration policies, Worlds, canvases, claimable plots, and shared sessions.
- Own all application screens, backend services, real-time sockets, public API integrations, databases, storage, deployment, and infrastructure across Scene Studio, MANGOs School, and the core MANGOs identity ecosystem.
- Designed the software boundary between web platforms and collaborator-produced course content, 3D scenes/assets, and Unity metaverses, enabling independently authored content to participate in a shared authenticated ecosystem.

SELECTED PUBLICATIONS

1. **M. B. Bo** et al., "Authentication and Identification System for Interconnected Metaverses: Consideration and Implementation," IEEE ECTI-CON, 2025. doi:10.1109/ECTI-CON64996.2025.11100953
   Designed and implemented identity management and interoperability across participating virtual environments.
2. **M. B. Bo** et al., "Design and Implementation of Tokenomics and Ecosystem Model for Interconnected Metaverses," IEEE CCISP, 2025, pp. 225-229. doi:10.1109/CCISP67522.2025.11282105
   Designed and implemented an ecosystem framework for participation, incentives, and value exchange.

SELECTED AWARDS

- **First Prize, APT IoT Smart Challenge for Agri-Tech** (2022)
  Developed AWS-backed cloud infrastructure and a deep learning component for an Arduino- and microcontroller-integrated smart greenhouse system.
- **Merit Award, Asia Pacific ICT Alliance Awards** (2023)
  Led a two-stage deep learning system for lung segmentation and pulmonary tuberculosis detection from chest X-rays, integrated into a doctor-patient cloud ecosystem. Official winner list.
- **Third Prize, Huawei Developer Competition Asia Pacific** (2024)
  Developed a blockchain-powered academic publishing platform on Huawei Cloud for research publishing, academic integrity, reputation, and revenue workflows. Competition result.

CERTIFICATIONS & LANGUAGES

Certifications: DevOps on AWS Specialization; Deep Learning Specialization; DeepLearning.AI TensorFlow Developer Professional Certificate; TensorFlow: Advanced Techniques Specialization.

Languages: Burmese (native); English (IELTS 8, official test; Diploma in English, Wall Street English).